Decent Coaching Classes

X Semi

Mind Your Math VIII

ANSWER KEY **Marks: 30**

Q.1 (A) **(4)**

(1) (C) $3x + 2y = 7$ (2) (D) 1 (3) (B) −3 (4) (A) −3

Hints: (2) $b^2 - 4ac = 25 - 24 = 1$. (3) $9 - 12 = -3$. (4) $D = 2(-1) - 1(1) = -3$.

Q.1 (B) **(4)**

(1) $d = 7 - 3 = 4$, so next term = $11 + 4 =$ **15**

(2) $3x^2 = 5x - 2$, so $\mathbf{3x^2 - 5x + 2 = 0}$

(3) $3x = 4y - 12$, so $\mathbf{3x - 4y + 12 = 0}$

(4) $t_n = a + (n - 1)d$, so $t_3 = 7 + 2 \times 2 =$ **11**

Q.2 (A) (any two, 2 marks each) **(4)**

(1) $D = 3 \times (-1) - 1 \times \mathbf{1} = \mathbf{-4}$; $D_x = 5 \times (-1) - 3 \times \mathbf{1} = \mathbf{-8}$; $D_y = 3 \times \mathbf{3} - 1 \times 5 = \mathbf{4}$
$x = D_x/D = -8/-4 = \mathbf{2}$, $y = D_y/D = 4/-4 = \mathbf{-1}$

(2) $x^2 - 5x - \mathbf{2x} + 10 = 0$; $x(x - 5) - 2(\mathbf{x - 5}) = 0$; $(x - 5)(\mathbf{x - 2}) = 0$
$\therefore x = \mathbf{5}$ or $x = \mathbf{2}$

(3) $d = 5 - 2 = \mathbf{3}$; $t_{12} = 2 + (12 - 1) \times \mathbf{3} = 2 + \mathbf{33} = \mathbf{35}$

Q.2 (B) (any two, 2 marks each) **(4)**

(1) Adding the two equations: $3x = 12$, so $\mathbf{x = 4}$. Then $4 - y = 1$, so $\mathbf{y = 3}$. (Check: $2(4) + 3 = 11$ ✓)

(2) $x^2 - 9x + 20 = 0$, so $x^2 - 4x - 5x + 20 = 0$, so $x(x - 4) - 5(x - 4) = 0$, so $(x - 4)(x - 5) = 0$. $\therefore$ **x = 4 or x = 5**

(3) $a = 4$, $d = 3$, $n = 15$. $t_{15} = 4 + (15 - 1) \times 3 = 4 + 42 =$ **46**

Q.3 (A) (any one, 3 marks) **(3)**

(1) $a = 2$, $b = \mathbf{-5}$, $c = \mathbf{3}$; $b^2 - 4ac = \mathbf{25} - \mathbf{24} = \mathbf{1}$
$x = (5 \pm 1)/4$, i.e. $(\mathbf{5 \pm 1})/4$ $\therefore x = 6/4 = \mathbf{3/2}$ or $x = 4/4 = \mathbf{1}$

(2) $d = 9 - 5 = \mathbf{4}$; $S_{20} = 20/2\ [2 \times \mathbf{5} + 19 \times \mathbf{4}] = \mathbf{10} \times [\mathbf{86}] = \mathbf{860}$

Q.3 (B) (any two, 3 marks each) **(6)**

(1) $D = 2(-1) - 3(3) = -2 - 9 = -11$; $D_x = 13(-1) - 3(3) = -22$; $D_y = 2(3) - 13(3) = -33$. (2 marks)
$x = D_x/D = -22/-11 = \mathbf{2}$, $y = D_y/D = -33/-11 = \mathbf{3}$. (1 mark) Check: $2(2) + 3(3) = 13$ ✓

(2) $2x^2 - 7x + 3 = 0$: $2 \times 3 = 6$; factors with sum −7 are −6 and −1.
$2x^2 - 6x - x + 3 = 0$, so $2x(x - 3) - 1(x - 3) = 0$, so $(x - 3)(2x - 1) = 0$. $\therefore$ **x = 3 or x = 1/2**

(3) $a = 7$, $d = 4$, $t_n = 99$. $t_n = a + (n - 1)d$, so $99 = 7 + (n - 1) \times 4$, so $92 = 4(n - 1)$, so $n - 1 = 23$. $\therefore$ **n = 24 terms**

Q.4 (any one, 5 marks) **(5)**

(1) Let the tens digit be x and the units digit be y. Number = $10x + y$. $x + y = 9$ … (i) (1 mark)
Interchanged number = $10y + x$. $(10y + x) - (10x + y) = 27$, so $9y - 9x = 27$, so $y - x = 3$ … (ii) (2 marks)
Adding (i) and (ii): $2y = 12$, so $y = 6$; then $x = 9 - 6 = 3$. (1 mark)

Original number = 10(3) + 6 = **36**. (Check: interchanged number 63 = 36 + 27 ✓) (1 mark)

(2) Let the integers be x and x + 1. $x(x + 1) = 132$, so $x^2 + x - 132 = 0$. (2 marks)

$x^2 + 12x - 11x - 132 = 0$, so $x(x + 12) - 11(x + 12) = 0$, so $(x + 12)(x - 11) = 0$. (2 marks)

$x = -12$ (rejected, not positive) or $x = 11$. ∴ The integers are **11 and 12**. (1 mark)

(3) $a = 3$, $d = 4$. $t_{25} = a + (n - 1)d = 3 + 24 \times 4 = 3 + 96 = \mathbf{99}$. (2 marks)

$S_{25} = n/2\ (t_1 + t_n) = 25/2 \times (3 + 99) = 25/2 \times 102 = 25 \times 51 = \mathbf{1275}$. (3 marks)